

Course Title: Microprocessors and Assembly Language Lab (CSE-4504)
Department of Computer Science and Engineering (CSE)
Islamic University of Technology (IUT), Gazipur

Lab # 03

8086 String Display and Loop Instructions using Assembly Language in EMU8086.

Objective:

Understand 8086 string display and loop instructions using Assembly Language Program.

Theory:

- **String Display Instruction**

At first define the string to be displayed under DATA SEGMENT:

.DATA

test_string DB 'My first string', 0Dh, 0Ah, '\$'

Then, display the string in the command prompt as:

MOV AH, 9

LEA DX, test_string

INT 21h

- **Loop:** LOOP instruction is a combination of a decrement of CX (i.e., count register). In 8086, LOOP decrements CX and if CX is not equal to zero, it jumps to the address indicated by the label. If CX becomes a 0, the next sequential instruction executes.

Assembly Language Program Example for Loop:

Count-controlled LOOP to display a row of 50 stars ().*

org 100h

.DATA ; Data segment starts

.CODE ; Code segment starts

MAIN PROC

mov ax, @DATA

mov ds, ax

xor cx, cx ; reset the CX register

mov cx, 50

mov ah, 2

mov dl, '*'

top: int 21h

loop top

mov ah, 4ch ; equivalent function number of RETURN

int 21h ; Input-Output Interrupt

MAIN ENDP

END MAIN

RET

Tasks to do:

1. Write an assembly language program that stores a string in a variable. Now, display the string and using loop calculate frequency of use or appearance of a given character in that string and display that as an output.

Sample Input / Output:

Input in a String: *input_string DB 'We are IUT Students', 0Dh, 0Ah, '\$'*

Output at DL: *we are iut students*

Input: *e*

Output: *3*